

# A loss-based framework for coherent prediction of competing risks

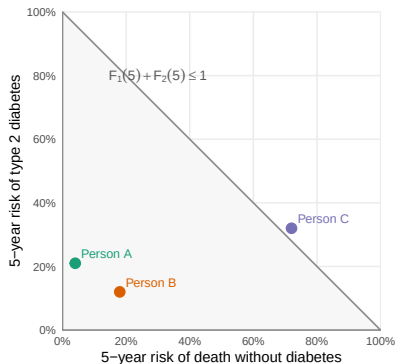
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# Competing risks risk prediction

$F_1(5 \mid x) = P(\text{type 2 diabetes by 5 years} \mid x),$

$F_2(5 \mid x) = P(\text{death without diabetes by 5 years} \mid x).$



# Motivation

When you estimate the cause specific risks at time  $\tau$  using semi-parametric Cox regression for the cause specific hazards, the sum of all cause specific risks can be more than 1 for certain covariate values.

We need to export the prediction model developed in protected environments, such that we can give it to the public.

It is unclear when to artificial censor subjects at the prediction time horizon.

## Setting: competing risks

Assume we are in a competing risks scenario and want to predict the risk of getting diagnosed with T2D and the risk of dying, at some fixed timepoint  $\tau$ . Denote:

- $T$  as the time between time0 and the date of an event
- $D \in \{1, 2\}$  as the cause of the event
- $X = (X^1, \dots, X^p)$  as a  $p$ -dimensional vector of baseline covariates, which can both be numeric or categorical.

Additionally we assume that the event time  $T$  is right censored at the random time  $C$ , where  $P(C > \tau | X) > 0$  almost surely. Denote  $\tilde{T} = \min(T, C)$  and  $\tilde{D} = \Delta D$  as the observed time and event, where  $\Delta = 1\{T \leq C\}$ .

The idea: fit both risks to the prediction horizon

Let  $\tilde{F}_j(t|x) = P(\tilde{T} \leq t, \tilde{D} = j|x)$  and  $o_{ij}(\tau) = 1(\tilde{D}_i = j, \tilde{T}_i \leq \tau)$

Brier-score loss function

$$BSE = \frac{1}{N} \sum_{i=1}^N \sum_{j=0}^2 \{\tilde{F}_j(\tau | X_i) - o_{ij}(\tau)\}^2.$$

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- One loss estimates the two cause-specific models **simultaneously**.

Here the sum starts at  $j = 0$  which is a method for handling right censoring ?.

## Focus of this talk: a single model, not prediction strategy

Learner library  $\longrightarrow$  K-fold cross-validation  $\longrightarrow$  Super Learner ensemble  $\longrightarrow$  external validation

Candidate learners can include cause-specific Cox models, direct binomial regression, boosting, neural networks, and constrained parametric hazard models.

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## Focus of this talk

One possible library member: a Cox-exponential model estimated by minimizing the Brier-score. A single learner that ensures simultaneous, coherent, and horizon-specific estimation, not the entire prediction strategy.

# The Cox-exponential model

For causes  $j \in \{0, 1, 2\}$ , assume constant baseline hazards:

$$\lambda_j(t \mid x) = \lambda_{0j} \exp(x^\top \beta_j), \quad \lambda_{0j} > 0.$$

Let  $\tilde{A}(x) = \sum_{j=0}^2 \lambda_{0j} \exp(x^\top \beta_j)$ . Then

$$\tilde{F}_j(\tau \mid x) = \frac{\lambda_{0j} \exp(x^\top \beta_j)}{\tilde{A}(x)} \{1 - \exp[-\tau \tilde{A}(x)]\}.$$

All parameter sets  $(\beta_0, \lambda_{00})$ ,  $(\beta_1, \lambda_{01})$  and  $(\beta_2, \lambda_{02})$  are estimated simultaneously by minimizing the Brier score loss.

## Brier score estimation

Assuming Cox-proportional hazards model and constant baseline hazards for all causes, our estimator becomes:

$$\hat{\theta}_{BSE} = \arg \min_{\theta} \frac{1}{N} \sum_{i=1}^N \sum_{j=0}^2 \left\{ \frac{\lambda_{0j} \exp(x_i^\top \beta_j)}{\tilde{A}(x_i)} \{1 - \exp[-\tau \tilde{A}(x_i)]\} - o_{ij}(\tau) \right\}^2.$$

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Cause specific risk prediction can then be obtained by substituting into  $F_j(\tau|x) = P(T \leq \tau, D = j|x)$ . Let

$$\hat{A}(x) = \sum_{j=1}^2 \hat{\lambda}_{0j} \exp(x^{\top} \hat{\beta}_j)$$

$$F_j(\tau|x) = \frac{\hat{\lambda}_{0j} \exp(x^{\top} \hat{\beta}_j)}{\hat{A}(x)} \{1 - \exp[-\tau \hat{A}(x)]\}, \text{ for } j \in \{1, 2\}$$

## Why estimate like this?

Estimating both cause-specific hazards this way:

- targets calibration and discrimination directly at the prediction horizon;
- performs well in non-proportional hazards data;
- prevents impossible pairs of predictions;
- produces a portable parametric learner for use outside protected environments.

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- prevents impossible pairs of predictions;
- produces a portable parametric learner for use outside protected environments.

Choosing a simple parametric model makes sense as we only fit the model for one specific timepoint, so we don't need the baseline hazard to vary over time.

## Artificial censoring at the prediction horizon

Artificial censoring replaces the observed follow-up by

$$\tilde{T}_i^{(5)} = \min(\tilde{T}_i, 5), \quad \tilde{D}_i^{(5)} = \tilde{D}_i 1\{\tilde{T}_i \leq 5\}.$$

It removes all event-time information after the five-year horizon. This affects the estimates of Cox models fitted using likelihood methods.

### It can lose predictive power

If proportional hazards is correct, later events inform the same  $\beta_j$  that governs hazards before five years. Discarding them wastes information.

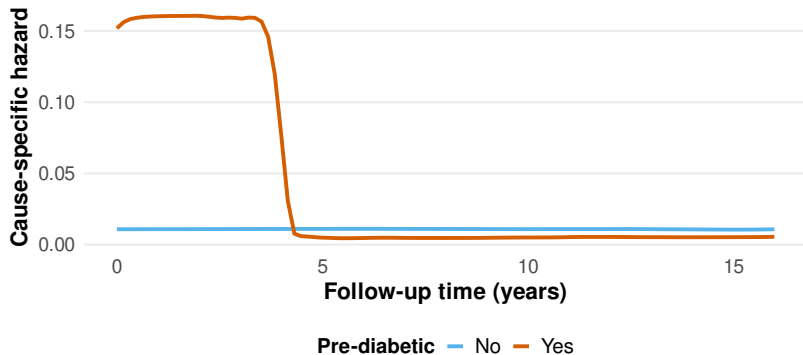
### It can gain predictive power

If effects change after five years, an untruncated Cox fit compromises across periods and can learn the wrong effect for the five-year target.

# Non-proportional hazards data

## Hazard of type 2 diabetes over follow-up

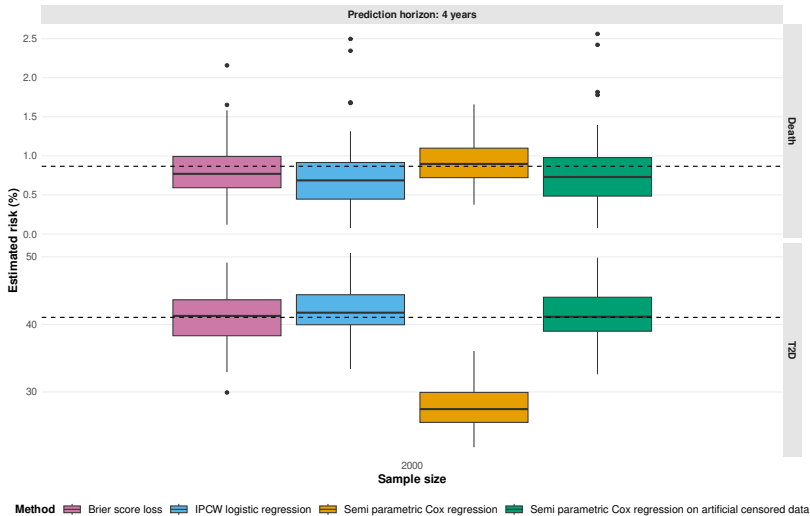
Hazard patterns differ by pre-diabetes status



# Simulation study

## Estimated versus true risks

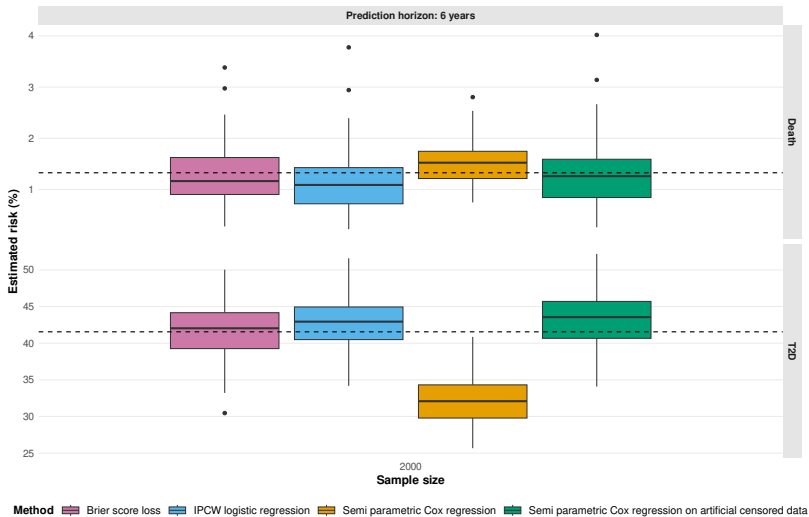
45-year-old male with pre-diabetes



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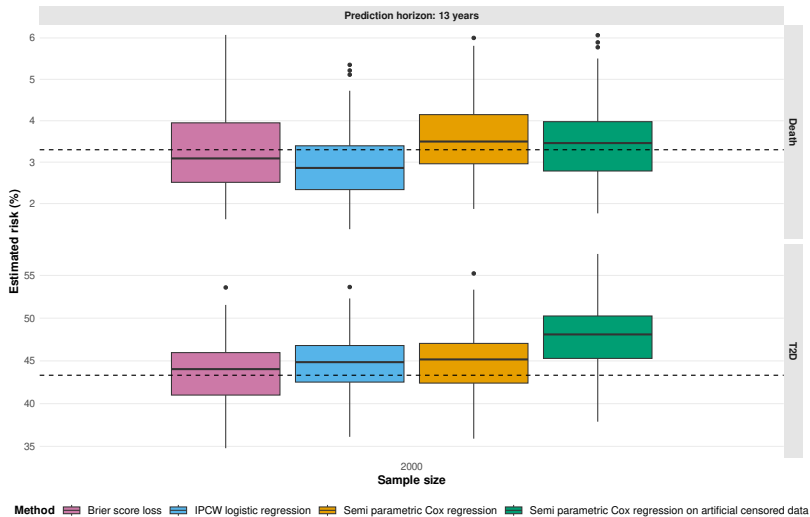
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- How to handle censoring; Use IPCW or fit a model for the censoring state?
- Need to extend framework to illness-death.